

IMPORTANT — FPL vs. LCEC Service Territory: Estero sits on the FPL/LCEC boundary. FPL serves parts of Estero; Lee County Electric Cooperative (LCEC) serves other areas. LCEC's hardening program is separate from FPL's and was significantly behind FPL's at the time of Hurricane Ian. Confirm which utility serves the specific substation and transformer assets identified in the vulnerability assessment before attributing FPL commitments to those assets.

1. Substations

Substations in flood zones face four primary failure modes: equipment inundation, water intrusion through conduit penetrations, buoyancy displacement of pad-mounted equipment, and corrosion from prolonged water contact.

Permanent Perimeter Floodwalls

The highest-confidence, most widely deployed solution. Reinforced concrete walls on driven sheet piles, sized to the design flood elevation plus freeboard.

- **Minimum industry standard:** 100-year flood elevation + 1 foot freeboard (Anixter/IEEE)
- **Con Edison post-Sandy standard:** 100-year flood + 3 feet, accounting for sea-level rise over 80-year lifespan
- **National Grid UK:** Hardened all 49 critical substations against 2050 climate-adjusted projections using concrete walls 1.6–2.4 m above grade
- **Soft defense alternative:** Earth bunds (compacted earthen berms) — lower ecological impact, site-specific height

Source: Con Edison Post-Sandy Enhancement Plan; Climate-ADAPT/EEA National Grid UK Case Study; Anixter Utility PowerUps

Equipment Elevation

FEMA P-348 identifies elevation as "the most effective overall solution" for primary electrical components. Transformers, switchgear, breakers, and control cabinets must be elevated above the design flood elevation on platforms, pedestals, or raised foundations. No electrical enclosure rated below IP67/NEMA 6 should be located below the design flood elevation (NEC/NFPA 70 requirement).

Source: FEMA P-348 (2017); FEMA P-2181 Fact Sheet 3.4.2

Watertight Control Buildings and Flood Doors

Dry floodproofing (sealing walls, penetrations, and door frames) works for floodwater up to approximately 3 feet depth. Beyond that, engineered perimeter walls are required. Con Edison installed over 270 watertight flood doors — types include reinforced aluminum slat doors, Kevlar fabric doors with lead-shot pressure-activated base seals, and roll-gate barriers.

Source: Con Edison Post-Sandy Enhancement Plan; T&D; World

Conduit Sealing

Con Edison installed 3,500+ expansive foam conduit seals at flood-level height. Roxtec sleeve systems seal conduit penetrations against hydrostatic pressure. Cable joint terminations at grade level are the highest

vulnerability in underground systems — sealed with metal housings, epoxy, or fiberglass boxes filled with water sealant.

Source: Con Edison Post-Sandy Enhancement Plan; Roxtec; Power & Cables

Deployable Barriers (Pre-Event)

- **AquaFence FloodWall:** Self-stabilizing using incoming water weight; FEMA-compliant; 500m perimeter deployed by 16 people in 4 hours — 375x more efficient than sandbags
- **Tiger Dam (water-filled bladder):** 1 unit = 500 sandbags; fills in 90 seconds; no heavy equipment needed
- **HESCO Floodline (earth-filled cells):** 1,500-sandbag equivalent in 20 minutes with a front-end loader

Source: AquaFence; US Flood Control; HESCO; IEEE Xplore — Power Grid Resilience Enhancement (2021)

FPL's Existing Substation Program (Baseline — Do Not Duplicate)

FPL has a formal Substation Storm Surge/Flood Mitigation Program under its FPSC-approved Storm Protection Plan. As of the 2026–2035 SPP, flood monitors are installed at 232 substations most susceptible to storm surge. The 2026–2035 plan (filed January 2025, approved April 2025) identifies additional substations requiring flood mitigation. In coastal/flood-prone areas where undergrounding isn't feasible, FPL places transformers on raised pads. Request FPL's current status for the specific Estero substation before recommending measures they've already implemented.

Source: FPSC 2026-2035 Storm Protection Plan filing; Gulfshore Business; FPL newsroom

2. Transformers — Vent Intrusion and Corrosion

Why This Is Happening

Transformers have three water intrusion pathways:

- **Conservator tank breathers** (oil-filled units): silica gel breathers draw in ambient air as oil cools, allowing moisture entry
- **Ventilation louvers** (dry-type units): openings required for airflow across windings and core
- **Cooling fin gasket interfaces:** degraded seals allow water entry between fins and tank

When water enters: dielectric oil strength drops, paper winding insulation saturates, internal corrosion accelerates, and electrical tracking paths form. Buoyancy forces can also physically displace pad-mounted units. The corrosion finding in Estero is consistent with repeated partial inundation — water entering through vents/fins but not fully submerging the unit, leaving concentrated mineral deposits as it evaporates.

Solutions by Severity (Most to Least Comprehensive)

Option A — Replace with IEEE Submersible Transformers [Utility-Scale Standard]

IEEE C57.12.23 (single-phase) and C57.12.24 (three-phase) define submersible distribution transformers specifically designed for continuous unattended operation while submerged under 3 meters (10 feet) of water. No ventilation openings — heat dissipates through the liquid-immersed design. Cast epoxy bushing assemblies eliminate gasketed penetrations. Con Edison installed 800+ submersible transformers and network protectors post-Sandy. This is the most justified long-term replacement for assets in confirmed flood zones.

Source: IEEE C57.12.23-2018; IEEE C57.12.24-2016; NEMA Storm Recovery Resources; Con Edison

Option B — Hermetically Sealed Oil-Filled Transformers

No conservator, no breather, no atmospheric exchange. Thermal expansion handled by corrugated fin walls that flex ± 50 kPa, with a nitrogen gas cushion at 0.02–0.05 MPa maintaining positive internal pressure — floodwater cannot enter even under submersion because internal pressure exceeds external hydrostatic pressure. Complete elimination of the intrusion pathway.

Source: IEEE C57.12 series; IEC 60076; Xbrele; Invexoil; Zetwerk

Option C — Totally Enclosed Non-Ventilated (TENV) Dry-Type Transformers

For dry-type replacements: TENV enclosures have no vent holes. Heat dissipates by radiation from the outer enclosure surface. Available in NEMA 4X (corrosion-resistant, spray-proof — specify Type 316 stainless steel for coastal/salt environments). **Note: NEMA 4X is not submersion-rated.** For submersion-possible locations, NEMA 6/6P or IP67/IP68 enclosures are required.

Source: Eaton; Hammond Power Solutions; Virginia Transformer Corp.

Option D — Cast-Resin Encapsulated (e.g., Siemens GEAFOL)

Windings physically potted in epoxy resin/quartz under vacuum — water cannot reach conductors even if the enclosure is breached. Air still circulates over the encapsulated winding exterior, so cooling is preserved. Requires an IP-rated enclosure for full flood protection. Standard GEAFOL rating is IP20 (open design); must be paired with IP67/68 housing for flood zones.

Source: Siemens GEAFOL Brochure; IEC 60076-11

Option E — Retrofit: GORE ePTFE Vent Membranes [Bridge Measure]

For situations where transformer replacement is not yet feasible. ePTFE microporous membranes installed over existing vent openings allow air/gas exchange (heat dissipation, pressure equalization) while blocking liquid water — pore size ~ 0.5 – 1 micron, smaller than the surface tension threshold for liquid water. GORE PolyVent XL is rated IP69K (highest rating — high-pressure hot water spray). **This is a bridge measure, not a permanent structural solution.**

Source: GORE Protective Vents; Electronic Specifier

Option F — Buoyancy-Actuated Louver Covers [Retrofit, Existing Units]

Louvered panels remain open during normal operation; internal floats rotate panels closed as floodwater rises — no electricity required. Wiper seals prevent bypass. Applicable as a retrofit to existing ventilated enclosures.

Source: Aquobex Floodbreak VSL; Floodbreak; USPTO Patent US11086109

3. Traffic Control Boxes

Traffic signal controller cabinets at grade level or on low pole sections are submerged or splash-damaged during flooding, causing signal failures with cascading public safety consequences — especially at evacuation routes and intersections near critical facilities.

Elevation — Primary Solution

FEMA P-2181 Fact Sheet 1.5 (the direct federal guidance on this topic): *"Raising ground-level and pole-mounted traffic signal controller cabinets above the estimated level of the design flood elevation will reduce contact with floodwaters, keeping the traffic signal working during and after a flood."*

- **Pole-mounted cabinets:** Detach and reattach higher on the pole (verify no DOT height restrictions)
- **Ground-level cabinets:** Raise and anchor on a concrete pedestal or steel post

- **Non-critical intersections:** 2 feet above BFE (USDOT FFRMS 2024)
- **Critical intersections** (evacuation routes, hospital access): **3 feet above BFE**
- **TxDOT cabinet extension:** Bolt-on base extension raises cabinet bottom above anticipated water — simple retrofit, no full replacement needed

Source: FEMA P-2181 Fact Sheet 1.5; USDOT Federal Flood Risk Management Standard Interim Guidelines (Dec. 2024); TxDOT Traffic Signals Manual

Enclosure Ratings

Rating	Protection Level	Use Case
NEMA 3R	Rain-tight outdoor — baseline	Standard (not flood-resistant)
NEMA 4X	Watertight spray + corrosion-resistant (Type 316 SS)	Coastal/salt-air environments; not submersion-rated
NEMA 6/6P	Temporary/prolonged submersion	Submersion-possible locations
IP67	Temporary submersion: 1m for 30 min	Flood-prone intersections — minimum spec
IP68	Continuous submersion beyond 1m	Underground junction boxes

Source: NEMA TS 2-2021; NEMA 250; IEC 60529; Miami-Dade County Section 685

Signal Heads

MUTCD minimum: 15 feet above pavement — inherently above most design flood elevations at urban intersections. Signal heads are typically not the failure point in flood events.

Source: MUTCD 11th Edition, Section 4D.09 (Dec. 2023)

Backup Power

- **Standard industry UPS:** 6–8 hours on lead-acid AGM batteries
- **Florida-tested solar + battery hybrid:** Provided power for several days without issues; cost range \$8,669–\$17,000 per installation (ITS deployment evaluation, 2022)
- **Solar-powered signals:** Fully grid-independent; DC/AC converter switches to battery within 50 milliseconds of power loss
- **Modern UPS cabinets:** Can include water/moisture sensors alerting maintenance during flooding

Source: ITS KRS Case Study 2022-sc00518; Miami-Dade Section 685; EnerSys ITS Traffic BBS; TxDOT UPS/BBU Manual

Flood Warning Integration

Commercial systems (ELTEC Corp, TAPCO) integrate high-water depth sensors with signal controllers to automatically trigger warning messages or signal changes when flood thresholds are reached. Deployed upstream of flood-prone intersections.

Source: ELTEC Corp; TAPCO High Water Warning System

4. FPL — What Has Already Been Done (Baseline)

Action	Status	Scope
Substation Storm Surge/Flood Mitigation Program	Active	232 flood monitors at most-at-risk substations
Storm Protection Plan (2023–2032)	FPSC-approved	\$14.5B total; ~\$7B undergrounding; 8 hardening programs
Storm Protection Plan (2026–2035)	Approved April 2025	Additional substations identified for flood mitigation
Lee County specific (2025)	Announced	5 critical feeder lines hardened; 855 mi vegetation trimming; pole inspections
Transformer elevation (coastal areas)	Ongoing practice	Transformers on raised pads where undergrounding not feasible
Underground lines (Lee County)	Active	Post-Ian Fort Myers Beach rebuild; 3,000+ SSUP projects systemwide
Underground performance	Documented	5–14x better than overhead during 2024 hurricane season
Coastal/flood area undergrounding	LIMITED	FPL acknowledges infeasible in most coastal/flood-prone areas due to groundwater

The 20-Year Question: FPL's Storm Secure Underground Program launched ~2007. Infrastructure built before that program would be 20+ years old and not hardened to current standards. The 2026–2035 SPP specifically addresses remaining un-hardened assets. To confirm Estero's specific substation last hardening date, request FPL's asset history through the FPSC filing or directly from FPL.

Source: FPSC filings 02358-2022, 03174-2025, 00290-2025; FPL newsroom; Gulfshore Business; CBS Miami

5. Writing This Into the Adaptation Plan

FEMA Justification — Community Lifelines Framework

FEMA's 2025 Local Mitigation Planning Policy Guide and 2025 Handbook establish Energy as Community Lifeline #1 — the upstream dependency for all other lifelines (transportation, water, communications). FEMA states: *"The best way to protect lifelines is to include them in state, local, tribal, or territorial mitigation plans."* **Plan inclusion is a formal prerequisite for BRIC and HMGP grant eligibility** — energy measures must appear in the LMS before grant applications can reference them.

Source: FEMA Local Mitigation Planning Policy Guide FP-206-21-0002 (April 2025); FEMA Local Mitigation Planning Handbook (June 2025)

Funding Pathways

- **BRIC** (Building Resilient Infrastructure and Communities): 75% federal / 25% local match. Explicitly funds substation hardening, utility undergrounding, microgrids, solar+battery. \$1.35B available FY2024.
- **HMGP** (Hazard Mitigation Grant Program): Permanent generators eligible when protecting a critical facility. Precedent: Polk County received \$1.44M for 31 generators at fire/EMS stations (2021).

- **DOE Grid Resilience State/Tribal Formula Grants** (IIJA): Formula grants for grid resilience against extreme weather. Florida is an active recipient.
- **FDEM Electric Grid Enhancement and Resiliency Grant**: State-level program; eligible categories include microgrids, battery storage, adaptive protective technologies (FLISR/SCADA), facility hardening, underground cable replacement. Presented directly to county LMS Working Groups in 2024.

Source: FEMA BRIC; FEMA HMGP; DOE Grid Resilience Program; FDEM EGER Grant (Osceola County LMS, March 2024)

BCA Requirement — Every Recommendation Must Justify $BCR \geq 1.0$

Per FEMA requirements, all HMA-funded projects must achieve a benefit-cost ratio of 1.0 or greater. Each recommendation must answer:

- **What** — specific system (e.g., submersible transformers per IEEE C57.12.23, not just 'replace transformers')
- **Where** — specific asset location tied to the vulnerability assessment finding
- **Why** — failure mechanism identified (vent intrusion → corrosion); hazard exposure confirmed (flood zone, documented inundation history)
- **Benefit** — link to a critical facility wherever possible (hospital, fire station, EOC, water treatment); document prior outage duration and cascading impacts
- **Funding** — identify the program before submission; BRIC requires plan alignment at application

Source: FEMA BCA Toolkit; FEMA Cost-Effectiveness and BCA Technical Document FY23; FEMA P-1019

Priority Recommendations Summary

Asset	Finding	Recommended Action	Standard / Justification
Substation	Confirmed flood zone	Perimeter floodwall (concrete on sheet piles) to 100-yr flood + 2–3 ft freeboard	Con Edison model; FEMA P-348; ASCE 24
Substation	Confirmed flood zone	Flood monitors + pre-positioned deployable barriers (AquaFence/Tiger Dam) if permanent wall infeasible short-term	IEEE stochastic framework; DOE 2024
Transformers	Vent corrosion from partial inundation	Replace with IEEE C57.12.23/C57.12.24 submersible units as assets reach end of life; interim: GORE ePTFE vent membranes	IEEE C57.12.23-2018; Con Edison precedent
Traffic control boxes	Flood-vulnerable at grade	Elevate to BFE + 2 ft (BFE + 3 ft at critical intersections) on concrete pedestals; specify NEMA 4X or IP67 enclosures; add 8-hr UPS or solar backup	FEMA P-2181 FS 1.5; USDOT FFRMS 2024

Asset	Finding	Recommended Action	Standard / Justification
All energy assets	Grid-dependent critical facilities	Identify downstream critical facilities (hospitals, EOC, water treatment); prioritize microgrids or permanent generators at highest-consequence nodes	FEMA HMGP; BRIC; FDEM EGER Grant

All findings adversarially verified across a minimum of 2 independent sources. | June 2026